# General Geospatial Inference with a Geographically Weighted Foundation Model

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## 1. Data and Task Design

We evaluate geographic representations using hierarchical, multimodal synthetic data (Fig. 1). Spatial units, fields, attributes, and targets are generated sequentially before measurement error and missingness are added. The primary task is to infer two health-burden indicators for held-out regions; attribute interpolation provides an auxiliary downstream evaluation. Data generation uses seed 20262001 and no external or real-world records.

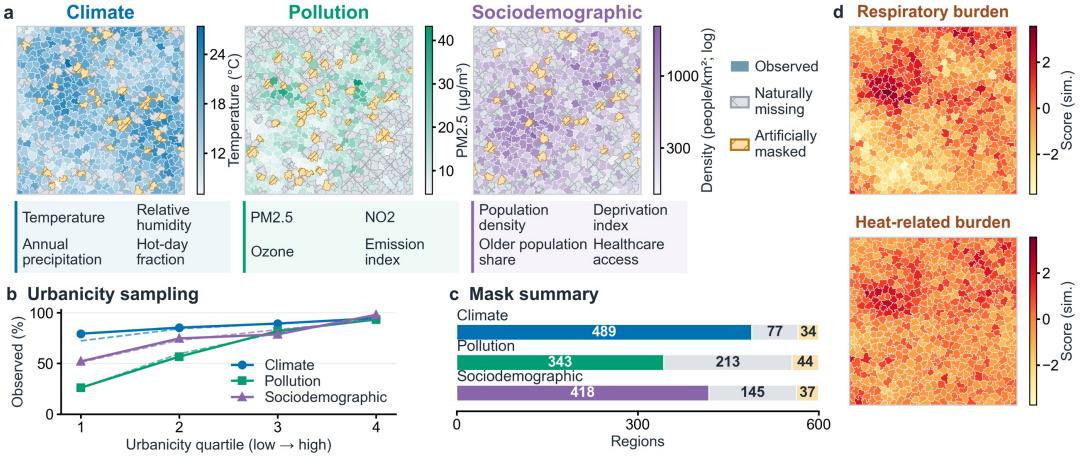



Fig. 1. Multimodal distributions and missingness. (a) Attribute and mask maps; (b) observed proportions by urbanization quartile; (c) regional counts by modality: observed, naturally missing, or artificially masked in a masked view; (d) health-burden target maps. Each modality's four attributes share a visibility mask. The artificial masks shown are illustrative; masks are resampled during self-supervised learning.

**(1) Spatial units and study area.** Seed points sampled according to simulated settlement density form 14,400 irregular atomic Voronoi polygons, merged along shared edges into 4,800 subregions and 600 regions. Units are internally connected; each region contains 2–16 subregions, each with one parent region. Coordinates $[0,1]^2$ represent a 200 km × 200 km Euclidean plane. Containment is retained; same-scale units sharing an edge are neighbors for spatial-autocorrelation calculations.

**(2) Spatial fields and cross-modal relationships.** Three independent fields each use 9,409 Gaussian kernels, centered on a regular 97 × 97 grid over $[-0.35, 1.35]^2$. Field peaks emerge from seeded standard-normal coefficients; their locations are not prescribed. Base bandwidths (0.128, 0.064, 0.192) are multiplied by $0.65 + 0.70g(s)$, where $g(s)$ varies smoothly from 0 to 1 across space. This positive multiplier widens kernels as g increases, producing smoother, longer-range spatial variation. Mixing weights combine the fields into modality-specific factors, creating cross-modal relationships that vary by location. For example, the pollution weight $0.65 - 1.10g(s)$ reverses the sign of the first field's contribution when it becomes negative. Each mixing vector is L2-normalized to prevent changes in coefficient magnitude from inflating variance. Separate means, amplitudes, and a shared background then control local levels, variability, and common structure.

**(3) Multimodal attributes and aggregation.** Latent factors, population density, and local perturbations generate three modalities (climate, pollution, and sociodemographic) with four attributes each. The attributes use linear, exponential, or logistic transformations to reflect their value ranges. The synthetic population totals 19.2 million; standardized log population density defines an urbanicity proxy representing population concentration. Before measurement error is added, regional climate and pollution attributes are area-weighted subregional means; social indices are population-weighted. Regional density equals total population divided by total area. These variables are simplified simulations; attribute-specific empirical distributions are not explicitly modeled. Full definitions and transformation rules are provided in the README.

**(4) Inference targets.** Regional respiratory- and heat-related burdens combine aggregated subregional environmental exposure and social vulnerability, spatially varying coefficients, neighborhood effects, and noise. Complete, measurement-error-free subregional temperature T, $PM_{2.5}$, deprivation D, and older-adult proportion A define heat exposure h, pollution p, and vulnerability v. Regional means are population-weighted. The weighted mean of subregional h × v defines the interaction $q_r$; the population-weighted standard deviation of p, $s_r$, measures within-region pollution variation. The regional targets are:

$$h_j = \frac{T_j - 18}{4},\quad p_j = \frac{ln(PM_{2.5,j}) - 2.4}{0.4},\quad v_j = \frac{D_j - 0.4}{0.2} + 0.4\,\frac{A_j - 0.18}{0.08} \tag{1}$$

$$y_{resp,r} = a_{1r}\bar{h}_r + a_{2r}\bar{p}_r + a_{3r}\bar{v}_r + 0.20(W\bar{p})_r + 0.15U_r + \varepsilon_{resp,r} \tag{2}$$

$$y_{heat,r} = a_{4r}q_r + a_{2r}\bar{p}_r + a_{5r}s_r + 0.15(W\bar{h})_r + 0.15U_r + \varepsilon_{heat,r} \tag{3}$$

Overbars denote population-weighted means; U is a smooth random spatial background. W aggregates neighbors using a Gaussian kernel of scale 0.09 (18 km), excluding self-links and normalizing rows. Coefficient baselines 0.60, 0.60, 0.40, 0.50, and 0.35 are each multiplied by $1 + 0.65\tanh f_k(r)$, where the $f_k$ are independent smooth random fields. Both targets include independent Gaussian noise with standard deviation $0.35[0.75 + 0.50g(r)]$, reusing the attribute-generation

transition g. Targets are noisy synthetic continuous scores, not measured disease prevalence. Regional target inference is primary; masked reconstruction of observed attributes tests information retention.

**(5) Local variation and measurement error.** Before attribute transformations, region-shared and subregion-independent perturbations are added, then the original means and standard deviations are restored. Gaussian measurement errors are generated separately at both scales before link-function transformations: standard deviations are 0.06 for regions and 0.08 for subregions, halved for precipitation and pollution; population density uses 0.10. Regional targets use attributes before measurement error is added.

**(6) Structured missingness.** Unequal observation resources are represented by the natural-missingness probability $sigmoid(a_m - b_m u)$, where m denotes modality and the urbanicity proxy u is standardized log population density. Climate, pollution, and sociodemographic $b_m$ values are 0.8, 1.8, and 1.1. Intercepts $a_m$ control overall missingness and are calibrated independently at each scale for expected rates of 15%, 35%, and 25%; actual regional rates are 12.8%, 35.5%, and 24.2%. Each modality's four attributes are missing jointly, with zero placeholders and visibility masks. Additional training masks provide verifiable reconstruction labels by masking only observed values.

## 2. From Heterogeneity to GWFM

Heterogeneity is a longstanding challenge in geographic modeling [1]. Despite spatial autocorrelation induced by Gaussian kernels, marked spatial and modality heterogeneity (Fig. 2) motivates location- and modality-specific representations.

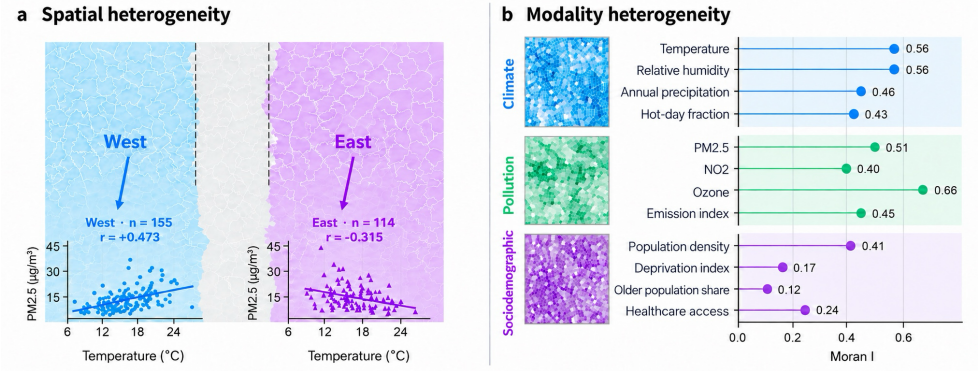



Fig. 2. Spatial and modality heterogeneity. (a) Spatial process heterogeneity: temperature–$PM_{2.5}$ associations differ between western ($x < 0.4$) and eastern ($x > 0.6$) zones defined by preset coordinate rules. (b) Modality heterogeneity: spatial patterns and clustering differ across modalities. Map and scatter-plot locations are schematic; simulated attributes' Moran's I [2] ranges from 0.124 to 0.656, indicating differing spatial autocorrelation across modalities.

**(1) Overall Framework.** Building on these ideas and PDFM [3], GWFM (Fig. 3) draws on location-dependent weighting in geographically weighted regression (GWR) [4]. Attribute and spatial-heterogeneity encoders separately learn attribute representations and geographic weights, which are combined into location-specific representations. Both use hierarchical spatial graphs and modality-heterogeneity-aware modules to integrate spatial context with shared and modality-specific information; prototype learning helps capture contextual differences between locations. Self-supervised pretraining combines masked reconstruction [5] and prototype alignment. A lightweight MLP predicts downstream targets from frozen representations.

**(2) Spatial context modeling and geographic weighting.** In GWFM, we decouple attribute representation from geographic weighting to characterize differences across locations using hierarchical and neighborhood information. Parent–child containment and visible neighbors define a hierarchical spatial graph. Both branches use GraphSAGE [6] with mean, max, and sum aggregation to learn context, producing attribute representations $h_k$ and geographic weights β. Their elementwise product modulates the same attributes according to context, yielding location-specific representations. X denotes visible inputs, G the hierarchical graph, and $E_k$ and $E_\beta$ the complete branch mappings:

$$h_{k,i} = E_0(X,G)_i, \quad \beta_i = E_\beta(X,G)_i, \quad z_i = \beta_i \odot h_{k,i} \tag{4}$$

**(3) Modality heterogeneity and awareness.** Both encoders combine shared and modality-specific mappings to capture cross-modal complementarity and semantic differences [7]. Shared mappings extract commonalities from available modalities at each node and in its spatial context. Specific mappings combine modality features, the shared component, and visibility masks to preserve modality characteristics. Shared and specific components are summed within modality-specific blocks, which are then concatenated to integrate sources. Omitting branch indices, $t_i^m$ denotes node i's modality-m features, $s_i$ the shared component, $v_i$ the visibility mask, and $f_m$ the modality-specific mapping:

$$u_i^m = s_i + f_m(t_i^m, s_i, v_i), \quad u_i = Concat(u_i^1, \ldots, u_i^M) \tag{5}$$

**(4) Prototype learning.** Learnable prototypes in the spatial-heterogeneity branch characterize and distinguish latent spatial pattern types across locations. Following DINO's cross-view consistency principle [8], student assignments from two independently masked views of each node are aligned with teacher assignments from originally observed inputs. This stabilizes assignments under incomplete observations and reduces dependence on particular observed modality combinations. Teacher parameters are updated by an exponential moving average; teacher outputs are detached, and prototype-similarity variance regularization discourages collapse. These are pretraining-only constraints. Here $p_i^{(1)}$ and $p_i^{(2)}$

are student assignments, $q_i$ teacher assignments, CE cross-entropy, and sg stop-gradient. Angle brackets denote averages over valid nodes within each scale, with equal scale weights:

$$\mathcal{L}_{proto} = \langle CE\big(sg(q_i), p_i^{(1)}\big) + CE\big(sg(q_i), p_i^{(2)}\big)\rangle/2 \quad (6)$$

**(5) Masking and reconstruction loss.** Independently masking originally observed modalities creates two training views. Modality-specific decoders reconstruct attributes from corresponding geographically weighted representation blocks. The loss uses only artificially masked observations, never naturally missing entries or placeholder values. Huber reconstruction loss [9] limits gradients from large residuals in long-tailed attributes. It is averaged over valid modalities and both views within each scale, then summed across scales. $\mathcal{L}_{\text{var}}$ is the variance regularizer on prototype similarities: penalizing insufficient variation in nodes' similarities to a given prototype preserves between-node differences and discourages collapse [10]. Total loss combines masked reconstruction, prototype alignment, and variance regularization, with weights $\lambda_p$ and $\lambda_v$, to jointly capture attributes, cross-modal complementarity, and differences across locations:

$$\mathcal{L} = \mathcal{L}_{mask} + \lambda_p \mathcal{L}_{proto} + \lambda_v \mathcal{L}_{var} \quad (7)$$

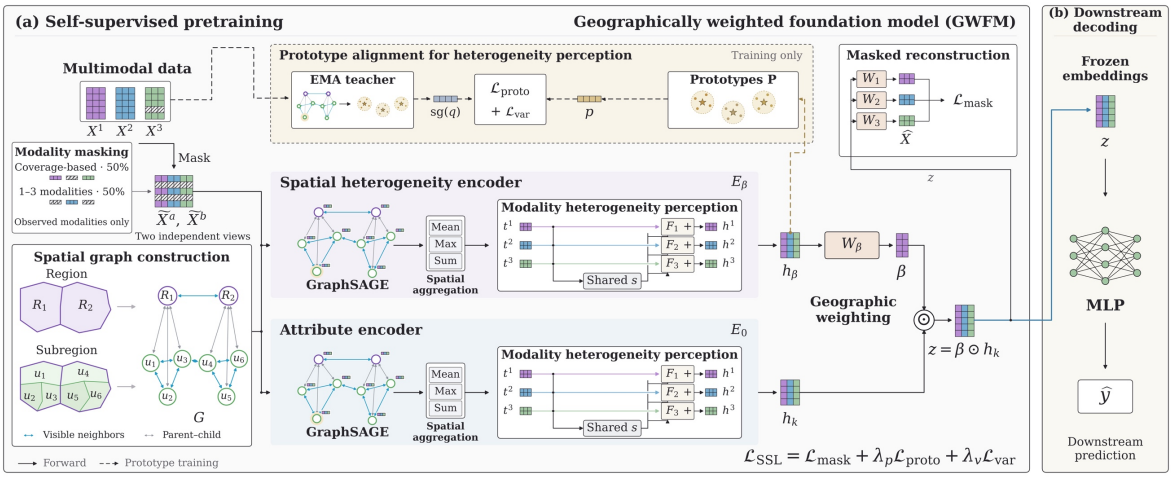



Fig. 3. GWFM. (a) Dual-branch encoding produces heterogeneity-aware geographic embeddings $\beta \odot h_k$ and reconstructs masked attributes in three modalities. (b) Frozen embeddings and a lightweight MLP predict downstream targets.

## 3. Experimental Setup

**(1) Data split.** Stratified sampling across 10 spatial groups divided 600 regions into 420/60/120 training/validation/test regions (42/6/12 per group; seed 1729); subregions inherited their parent regions' split assignments. Graphs were split-specific. Only training-split data updated pretraining parameters; test attributes did not contribute to updates. Standardization used training observations or labels. For inference, frozen encoders used all naturally observed attributes. For interpolation, the target modality was excluded from all regional inputs before encoding, while other modalities and naturally observed subregional attributes remained available. Predictors were fitted to training labels and tuned on validation data. Coordinate baselines used only training-region health labels for inference and observed training-region attributes for interpolation. All methods used the same data splits and were evaluated on the same regional samples for each target.

**(2) GWFM settings.** *① Pretraining:* GWFM used PyTorch Geometric (PyG) [30]. Both branches used single-layer GraphSAGE (hidden width 64; 48-dimensional embeddings, 3×16; 8 neighbors); GWFM had 106,297 parameters. In each of two views, each node independently sampled a mask according to empirical missingness rates with 50% probability or randomly masked 1–3 modalities otherwise. Validation masks were fixed at a 30% masking probability. Huber loss used δ=1; prototype-consistency weight $\lambda_p$ and variance-regularization weight $\lambda_v$ were both 0.05, with 16 prototypes and teacher EMA decay 0.99. Training used seeds 11/23/37, full-batch Adam [11], learning rate 0.001, 300 epochs with cosine annealing [12], and gradient-norm clipping at 5. Standardization used training data; the encoder minimizing validation reconstruction loss was frozen. Within each split and modality, nodes connected to ≤8 nearest visible same-scale neighbors, excluding themselves. Messages used distance-decay weights $w_{j\to i} = \exp\left[-\min\left\{(d_{ij}^2 - d_{i,min}^2)/b^2, 30\right\}\right]$, where $d_{ij}$ is Euclidean distance, $d_{i,min}$ the nearest visible-neighbor distance for that modality, and $b$ a branch-, scale-, and modality-specific learned bandwidth. The exponent magnitude was capped at 30 for stability; containment defined bidirectional parent–child edges with weight 1. *② Downstream predictors:* Two-layer GELU [13] MLPs used the full 48-dimensional embedding or individual 16-dimensional modality blocks. Each head selected hidden width {32,64,128} and weight decay {0.0001,0.01} independently by validation MSE after standardizing each target using training-label statistics. AdamW [14] used learning rate 0.003, ≤500 epochs, early-stopping patience 40, and decoder seed 11. Final settings appear in the code and tuning records.

**(3) Baselines.** Coordinate-based comparators were K-nearest-neighbor regression (KNN) [15], inverse-distance weighting (IDW) [16], and Gaussian process regression (GPR) [17]; regional target inference used only the 420 training regions' labels. Task-adapted PDFM [3] was trained from scratch. ① Coordinate-based KNN used k=8 for both targets. ② IDW weighted neighboring labels by inverse-distance powers: 32 neighbors/power 2 for respiratory burden; 16 neighbors/power

1 for heat burden. ③ GPR kernels selected by validation MSE were Matérn (ν=1.5) for respiratory burden and RBF for heat burden. ④ PDFM used single-layer GraphSAGE with normalized distance-weighted mean aggregation at both scales and Huber loss for observed-attribute reconstruction; hidden width was 64, embedding dimension 48 (16 per modality), and parameter count 16,764. Regional Gaussian edges used fixed bandwidth $b \approx 14.402$ km and weights $w_{j\to i} = \exp(-d_{ij}^2/b^2)$, retaining edges satisfying $w_{j\to i} \geq 0.1$. Per modality, subregions connected to ≤8 currently visible nearest neighbors, excluding themselves, with relative inverse-square weights $w_{j\to i} = d_{i,min}^2/d_{ij}^2$; $d_{ij}$ is Euclidean distance and $d_{i,min}$ the nearest visible-neighbor distance for that modality. Bidirectional parent–child edges had weight 1. During self-supervised pretraining, PDFM followed GWFM's masking and reconstruction strategy and checkpoint-selection procedure, but used a single masked view. Training seeds and downstream settings matched those of GWFM.

**(4) Common evaluation settings.** $R^2$ was calculated for each attribute or regional target, then averaged with equal weights. Neural results are three-seed means with sample standard deviations; coordinate methods use fixed runs.

## 4. Results and Analysis

Regional target inference was the primary task; interpolation, ablations, and visualization assessed heterogeneity modeling and performance gains. Interpolation also tested modality-specific information retention.

Table 1. Geospatial inference and interpolation $R^2$. Modal 1–3 are modality-specific blocks of the full embedding (climate, pollution, sociodemographic); Multi-Modal is the full representation. Neural results are averaged over three seeds.

| Inference (Primary task) | KNN | IDW | GPR | PDFM Modal-1 | PDFM Modal-2 | PDFM Modal-3 | PDFM Multi-Modal | GWFM Modal-1 | GWFM Modal-2 | GWFM Modal-3 | GWFM Multi-Modal |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Respiratory-related burden | 0.534 | 0.545 | 0.564 | 0.624 | 0.705 | 0.731 | 0.798 | 0.776 | 0.639 | 0.696 | 0.834 |
| Heat-related burden | 0.240 | 0.274 | 0.271 | 0.310 | 0.308 | 0.320 | 0.304 | 0.307 | 0.445 | 0.347 | 0.456 |
| Overall inference | 0.387 | 0.409 | 0.418 | 0.467 | 0.507 | 0.526 | 0.551 | 0.542 | 0.542 | 0.522 | 0.645 |
| **Interpolation (Auxiliary task)** | | | | | | | | | | | |
| Temperature | 0.545 | 0.541 | 0.529 | 0.842 | 0.751 | 0.771 | 0.830 | 0.975 | 0.723 | 0.745 | 0.967 |
| Relative humidity | 0.520 | 0.553 | 0.595 | 0.793 | 0.696 | 0.767 | 0.816 | 0.944 | 0.607 | 0.636 | 0.940 |
| Annual precipitation | 0.334 | 0.360 | 0.356 | 0.751 | 0.640 | 0.645 | 0.737 | 0.942 | 0.546 | 0.590 | 0.935 |
| Hot-day fraction | 0.247 | 0.398 | 0.460 | 0.854 | 0.835 | 0.829 | 0.842 | 0.957 | 0.755 | 0.755 | 0.956 |
| PM2.5 | 0.450 | 0.443 | 0.474 | 0.570 | 0.705 | 0.581 | 0.649 | 0.539 | 0.931 | 0.521 | 0.924 |
| NO2 | 0.322 | 0.360 | 0.404 | 0.652 | 0.751 | 0.714 | 0.767 | 0.634 | 0.947 | 0.639 | 0.946 |
| Ozone | 0.505 | 0.456 | 0.593 | 0.681 | 0.795 | 0.725 | 0.779 | 0.590 | 0.923 | 0.554 | 0.915 |
| Emission index | 0.353 | 0.419 | 0.435 | 0.566 | 0.741 | 0.615 | 0.725 | 0.511 | 0.922 | 0.387 | 0.910 |
| Population density | 0.345 | 0.360 | 0.417 | 0.561 | 0.582 | 0.665 | 0.639 | 0.590 | 0.594 | 0.887 | 0.882 |
| Deprivation index | 0.123 | 0.156 | 0.123 | 0.571 | 0.562 | 0.660 | 0.676 | 0.520 | 0.355 | 0.898 | 0.883 |
| Older population share | 0.073 | 0.034 | -0.000 | 0.438 | 0.422 | 0.391 | 0.481 | 0.404 | 0.298 | 0.787 | 0.777 |
| Healthcare access | 0.252 | 0.249 | 0.264 | 0.635 | 0.616 | 0.663 | 0.692 | 0.461 | 0.557 | 0.869 | 0.858 |
| All 12 variables (mean) | 0.339 | 0.361 | 0.388 | 0.659 | 0.675 | 0.669 | 0.719 | 0.672 | 0.680 | 0.689 | 0.908 |

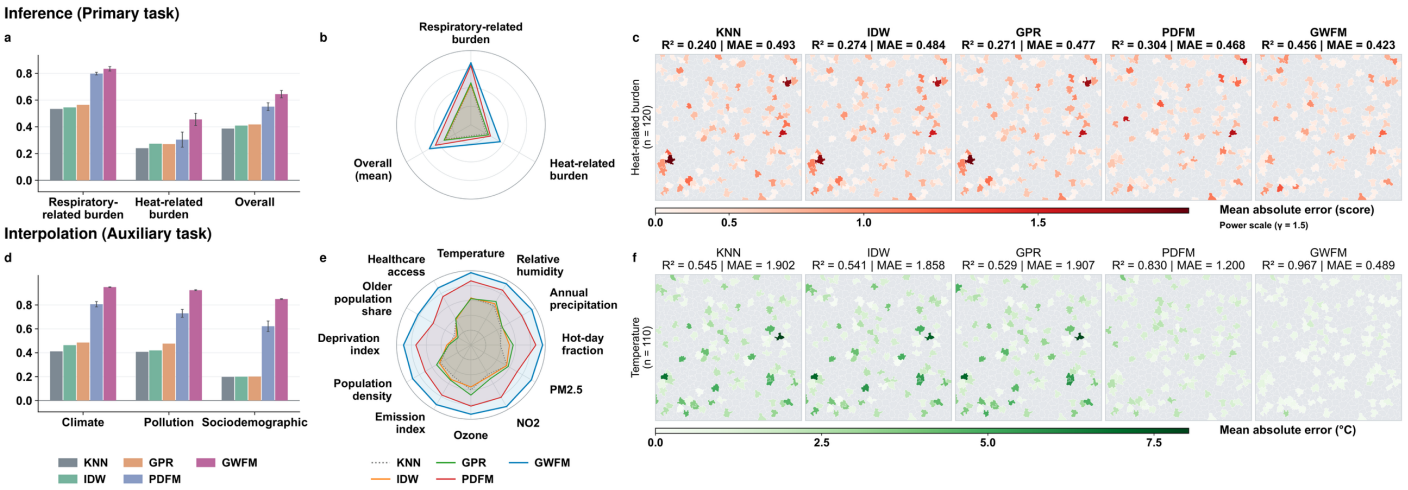



Fig. 4. Regional target inference and interpolation. (a–b) Regional target inference $R^2$; (c) heat-burden absolute-error maps; (d–e) interpolation $R^2$ by modality and attribute; (f) temperature-interpolation absolute-error maps.

| a Inference (Primary task) | w/o Geographic weighting | w/o Modality perception | w/o Prototype regularization | GWFM (Full) |
|---|---|---|---|---|
| Respiratory-related burden | 0.815 | 0.801 | 0.829 | 0.834 |
| Heat-related burden | 0.447 | 0.451 | 0.439 | 0.456 |
| Overall inference | 0.631 | 0.626 | 0.634 | 0.645 |
| **Interpolation (Auxiliary task)** | | | | |
| Temperature | 0.958 | 0.949 | 0.965 | 0.967 |
| Relative humidity | 0.929 | 0.912 | 0.932 | 0.940 |
| Annual precipitation | 0.924 | 0.910 | 0.936 | 0.935 |
| Hot-day fraction | 0.951 | 0.948 | 0.952 | 0.956 |
| PM2.5 | 0.888 | 0.873 | 0.913 | 0.924 |
| NO2 | 0.923 | 0.920 | 0.935 | 0.946 |
| Ozone | 0.902 | 0.890 | 0.907 | 0.915 |
| Emission index | 0.871 | 0.856 | 0.888 | 0.910 |
| Population density | 0.861 | 0.869 | 0.877 | 0.882 |
| Deprivation index | 0.867 | 0.858 | 0.862 | 0.883 |
| Older population share | 0.757 | 0.738 | 0.770 | 0.777 |
| Healthcare access | 0.845 | 0.838 | 0.848 | 0.858 |
| All 12 variables (mean) | 0.890 | 0.880 | 0.899 | 0.908 |

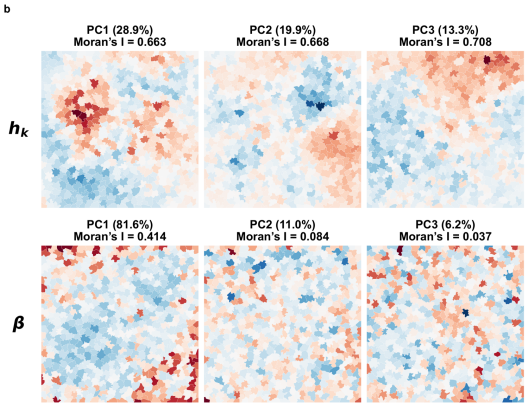



Fig. 5. Ablations and representations. (a) Geographic-weighting, modality-awareness, and prototype-learning ablations; (b) first three principal components, explained variance, and Moran's I of attribute representations $h_k$ and weights β (seed 11).

**(1) Overall performance and modality awareness.** GWFM improved mean $R^2$ over task-adapted PDFM by 17.0% for inference and 26.2% for interpolation (Table 1). Coordinate baselines struggled with multimodal attributes and their combinations; GPR had the highest mean $R^2$ among these baselines, possibly because spatial covariance modeling suits the smooth background. PDFM's high-dimensional embeddings improve representation quality but offer limited explicit modeling of spatial and modality heterogeneity. Modality blocks generally performed better on corresponding attributes, while full representations performed better overall, supporting shared–specific complementarity.
**(2) Spatial heterogeneity and location-specific modulation.** GWFM reduced errors across multiple regions: MAE fell approximately 9.7% for heat-related burden and 59.2% for temperature relative to PDFM (Fig. 4). Nearby locations may be correlated yet differ in attribute combinations and mappings, which uniform mean aggregation may inadequately capture. Separately learned attribute representations and geographic weights enable context-dependent modulation, potentially reducing over-smoothing of local differences and preserving location specificity.
**(3) Component contributions and representations.** Removing modality-heterogeneity awareness caused the largest overall decline, especially in interpolation; removing prototype learning or geographic weighting also impaired some tasks (Fig. 5). For the first three principal components, PCA [18] showed stronger spatial clustering of attribute representations and weaker spatial autocorrelation of geographic weights, consistent with preserving attribute structure and modulating local differences, respectively. Ablations and visualizations support that heterogeneity modeling matters for geospatial representation.

## 5. Limitations and Future Perspectives

By jointly modeling modality and spatial heterogeneity, GWFM performs well overall in geospatial inference and interpolation and helps interpret spatial representation differences. However, fusion does not outperform modality blocks on every attribute; sub-modality interpretation remains difficult. One synthetic dataset and a fixed split leave real-world applicability unverified. More complex heterogeneity and large contiguous gaps from constrained sampling may demand stronger cross-region inference. Within-group sampling tests neither spatial block holdouts nor cross-region extrapolation [19]. Future work should test these settings and extend to spatiotemporal learning and causal inference.
**(1) From static spatial to spatiotemporal representations.** Following PDFM [3], which combines spatial embeddings with TimesFM [20] forecasts, an external pretrained temporal model offers an initial baseline, although separate encoding may limit spatial–temporal coupling. Inspired by STMAE [21] and UniST [22], masking and reconstructing temporal segments of spatial embeddings across time slices could learn dynamic representations. Joint spatiotemporal graphs inspired by STFGNN [23] could capture coupled dependencies within one self-supervised framework. Irregular observations and modality differences in sampling frequency, temporal coverage, and input length remain challenging. Explicit timestamps, intervals, and observation masks, combined with continuous-time neural differential equations such as Neural CDE [24], could distinguish missingness and sampling differences from genuine dynamics.
**(2) From attribute representations to spatial causal inference.** This method learns predictive associations, not causal relationships. Given specified treatments and confounders, neural models can estimate treatment effects through representation balancing and counterfactual inference [25], subject to identification assumptions. Weather2vec [26], for example, encodes surrounding weather into location-level representations and uses propensity-score matching to assess the effects of installing emission controls at power plants on ozone pollution. Furthermore, spatial spillovers complicate inference: treatments may affect neighboring outcomes, requiring explicit modeling of local and neighboring treatments, not confounding adjustment alone [27]. Future work could use GCCM [28] and its spatiotemporal extension STCCM [29] for spatial and spatiotemporal state reconstruction and convergent cross mapping to explore potential variable–target causal directions, then test candidate relationships as soft priors in GWFM's variable-interaction and causal-decoding modules.

## 6. Reproducibility and AI Disclosure

**(1) Reproducing the experiments.** Code root: GWFM; Python 3.12.2; pinned dependencies: requirements-lock.txt. Data-generation settings: configs/dataset.json; model/ablation settings: configs/. Generation, splitting, and encoder training use fixed seeds. Install dependencies and run from the code root with a new, empty output directory:

```
python -m pip install -r requirements-lock.txt
python -m gwfm all --work-dir ./fresh_run
```

**(2) Reproducibility.** Data, frozen weights, embeddings, and predictions are provided. We reproduced GWFM, its three ablations, PDFM, and simple spatial baselines from scratch in a separate Conda environment. Neural encoders trained for 300 epochs used seeds 11, 23, and 37. Downstream MLPs were tuned on validation data and evaluated on the test set. Saved weights reproduce embeddings elementwise; training from scratch may vary slightly across hardware and floating-point environments. GitHub: https://github.com/wy-st/GWFM.
**(3) AI disclosure.** Codex and ChatGPT aided data synthesis, modeling, training/evaluation scripts, plotting, report editing, and refactoring. An AI image tool aided text/figure layout; Fig. 2a is schematic. Tables, error maps, and statistics use saved data and predictions. Using requirements, formulas, code, and results, AI checked formula–implementation consistency and missingness/splits, traced weights/predictions, and regenerated figures. Consistency checks used artifact recomputation and regression checks on statistics. README.md also discloses AI use. The author reviewed and revised AI-generated content and automated checks and takes responsibility for correctness, interpretation, and reproducibility.

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